Lecture 11: AWS Well-Architected Framework (WAP)

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Agenda

- 6 pillars for AWS Well-architected
- Operational Excellence
- Security
- Reliability
- Performance Efficiency
- Cost Optimization
- Sustainability

Overview



- https://medium.com/@vasanthabalaji/aws-well-architected-framework-6-pillars-explained-83af8e29ce0d

Operational Excellence

- The ability to run and monitor systems to deliver business value and continually improve supporting processes and procedures.
- Maintain reliability and efficiency

Operational Excellence

- Perform Operations as Code: Automate operations to make them repeatable and predictable by scripting.
- Annotate Documentation: Keep documentation up-to-date to help teams understand environment and operational practices.
- Make Frequent, small, reversible changes: Implement changes gradually to limit the impact of any errors.

Operational Excellence

- Anticipate Failure: Regularly execute failure scenarios to test responses and understand the impact.
- Learn from all operational failures: Analyze failures for root causes, share learnings, and use them to improve.
- Improve through game days: Simulate different failure modes in a controlled environment to validate the understanding and effectiveness of the environment's response.

Operational Excellence

- Monitoring and logging: Monitor health and performance using AWS CloudWatch and AWS CloudTrail.
- Automation of Processes: Automate responses to operational issues by using AWS Lambda and AWS Auto Scaling.
- Feedback loops: Implement mechanism to rapidly and safely deploy changes based on the real-world use of the system.

Security

- Ensure the integrity and confidentiality of information by protecting systems, data, and assets.
- Fundamental aspect of system architecture, particularly in cloud environment like AWS such as IAM, Security Group, NACL, VPC…

Security

- Implement a strong Identity Foundation: use AWS IAM to manage access securely.
- Enable Traceability: Monitor, alert, and audit actions and changes to the environment in real time using AWS CloudTrail and AWS Config.
- Apply Security at All layers: Implement security in every layer (e.g., edge network, VPC, application, operation system, and data layers).

Security

- Protect Data in Transit and at Rest: Encrypt data at rest using AWS KMS and in transit using SSL.
- Keep People Away from Data: Utilize automated tools to reduce the risk of human error and discuss the principle of least privilege.
- Prepare for Security Events: Introduce AWS tools and practices for incident response, including the AWS Incident Response whitepaper and Amazon GuardDuty for threat detection and monitoring.

Security

- Identity and Access Management (IAM): Detail practical steps for implementing IAM best practices, such as multi-factor authentication and fine-grained access controls.
- Data Encryption: Outline how to setup encryption for AWS services used to store or process data.
- Continuous Compliance Monitoring: Integrate AWS services that help maintain and enforce security standards automatically.

Reliability

- The ability of a system to recover from infrastructure or service disruptions, dynamically acquire computing resources to meet demand and mitigate disruptions.
- Maintain user trust and ensure continuous service delivery, especially in cloud environments.

Reliability

- Automatically Recover from Failure: Design systems that can automatically heal themselves.
- Test Recovery Procedures: Regularly check how a system recovers from failures using automated tests.
- Scale Horizontally to Increase Aggregate System Availability: Utilize Auto Scale mechanisms to achieve higher availability and reliability.

Reliability

- Setup monitoring and Alerts: Implement monitoring with AWS CloudWatch and set up alerts to respond to changes in system performance in real time.
- Manage Service Limits and Backup strategies: Outline strategies for managing service quotas and limits and implement robust data backup solutions.
- Implement Failover Strategies: Introduce concepts such as DNS failover, multi-region deployment, and the use of Amazon Route 53 and Elastic Load Balancing to ensure continuous operation.

Reliability

- Infrastructure Automation: Leverage Auto Scaling, Elastic Beanstalk, or CloudFormation for infrastructure provisioning and management.
- Dependency Management: Manage and mitigate risks from third-party services or resource bottlenecks.
- Regular Stress Testing: Regularly do stress tests and simulated outages to validate the system's resilience and recovery procedures.

Performance Efficiency

- The ability to use computing resources efficiently to meet system requirements and maintain that efficiency as demand changes and technologies evolve.
- It is critical to optimize both speed and cost-effectiveness, ensure applications run smoothly and economically.

Performance Efficiency

- Democratize Advanced Technolgies: AWS make it easier for users to adopt advanced technologies (like ML and IoT) without deep expertise.
- Go Global in Minutes: Deploy applications in multiple regions around the world to improve latency and reduce the impact of regional disruptions.
- Use Serverless Architectures: This architecture allows users to focus on their application code while cloud provider manages the run-time environment, thus optimizing performance.

Performance Efficiency

- Right resource: Choose the right resource types and sizes based on workload requirement, like EC2 or Lambda.
- Monitoring: Monitor the performance, using CloudWatch or X-Ray, to gain insights into how applications are performing in real time.
- Trade-off: Make the trade-off decision between different types of resources or services to optimize performance, such as choosing between compute optimized vs. Memory optimized resources.

Performance Efficiency

- Benchmarking: Understand the benchmark using AWS tools to measure the performance of applications under different configurations.
- Automated Scaling: Implement elastic scaling with AWS Auto Scaling and Elastic Load Balancing to handle changes in load dynamically.
- Optimize Data Storage: Optimize data storage and retrieval. Focus on choosing between S3, EBS, and RDS based on their performance characteristics.

Cost Optimization

- Understand and control where money is being spent.
- Select the most appropriate and right number of resource types.
- Analyze spend over time, and scale to meet business needs without overspending.
- Maximize the value of cloud investments and ensure financial sustainability.

Cost Optimization

- Adopt a Consumption Model: Pay only for the computing resources needed and scale up or down as required.
- Measure Overall Efficiency: Measure the business output of the service and the costs associated with delivering it, ensure that resources are being used efficiently.
- Stop Spending Money on Data Center Operations: Contrast cloud costs to traditional data center costs, focusing on how the cloud shifts capital expenditure to variable expenditure.

Cost Optimization

- Right Size: Analyze and modify instances to ensure to use efficient resource types and sizes.
- Increase Elasticity: Leverage cloud features like Auto Scaling to match supply with demand without over-provisioning.
- Choose the right pricing model: Utilize the purchasing options like Reversed Instances, Spot Instances, and Savings Plans to reduce costs.

Cost Optimization

- Cost-effective resources: Select the most cost-effective resources for different workloads (e.g., using Amazon EC2 Spot Instances for interruptible workloads).
- Monitor and Analyze Costs: Use AWS Cost Explorer and AWS Budgets to monitor, track, and analyze costs.
- Optimize Storage: Optimize costs related storage solutions, such as using Amazon S3 lifecycle policies to archive or delete unused data.

Sustainability

- Create and run applications in a way that minimizes the impact on the environment.
- Consider environmental impacts as part of the architectural and decision-making process.

Sustainability

- Understand impacts: Thoroughly assess the environmental impacts of IT resources.
- Maximize Utilization: use resources efficiently to not only reduce cost but also minimize unnecessary energy consumption.
- Reduce Downstream impacts: Choose resources and servers can affect the sustainability of operations, both directly and indirectly.

Sustainability

- Optimize Energy Usage: Optimize compute and storage resources to reduce energy consumption, such as selecting appropriate instance types and sizes.
- Increase Resource Efficiency: Increase efficiency through modernization and serverless architectures, which can reduce the number of servers in use and the associated energy consumption.
- Reuse and Recycle Resources: Recycle unused resources and use managed services to decrease idle computing times.

Sustainability

- Selection of Green Technologies: Choose AWS services and features that are designed to be more energy-efficient.
- Implement Performance Efficiency: Efficient design inherently supports sustainability by reducing the usage of resources needed.
- Use of Renewable Energy: Leverage AWS initiatives to achieve energy sustainability.

Conclusion

- Operational Excellence: Focus on running and monitoring systems to deliver business value and continually improving processes and procedures.
- Security: Involve protecting information, systems, and assets while delivering business values through risk assessments and mitigation strategies.
- Reliability: Ensure a workload performs its intended function correctly and consistently when it is expected.

Conclusion

- Performance efficiency: Focus on using computing resources efficiently to meet system requirements and maintaining that efficiency as demand changes and technologies evolve.
- Cost Optimization: Avoid unnecessary cost
- Sustainability: Focus on achieving business objects with least environmental impact.

Conclusion

- Adopt principles to reach the operational excellence.
- Review and enhance the security measures.
- Assess and improve the systems' reliability.
- Adjust the cloud infrastructure to enhance performance efficiency.
- Regular review and apply the best practices to usage and cost.
- Integrate sustainability considerations into the cloud strategy and decision-making process.

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